

Clinical Anatomy and Physiology of Birds

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INTRODUCTION

All birds kind of look the same and are easily recognizable as birds. It is because, for the most part, their anatomy and physiology are shaped and constrained by flight, a demanding form of locomotion, for which being aerodynamic with a small size and low weight is imperative. Birds have a very high metabolism (higher than any other animal group on the planet) and most organ systems function at physiological extremes (most organ systems are more efficient than their mammalian counterparts especially the respiratory and cardiovascular systems).

As birds are reptiles, some similarities are expected (especially with crocodilians). For most organ systems, birds can be considered the step further after reptiles. For instance, the scales become the feathers, the kidneys have added looped nephrons, the air sacs have extended further in the body with added functions, the coelomic cavities have been further divided, and they became homeothermic (like some dinosaurs before them). However, bird anatomy and physiology are more homogeneous than reptiles because of the constraints of homeothermy, increased metabolism, and flight. As such, reptiles present much more complex (and one must say fascinating) diversity in anatomy, physiology, and even general body plans (you'll see that in the first reptile lecture). It is also harder to learn reptile anatomy for this reason while bird anatomy is at the same time fascinating and similar across species, which makes it easier to remember.

A good knowledge of anatomy and physiology is paramount to clinical work, especially when considering anesthesia, surgery, imaging, interpretation of blood work (blood work is interpreted in the context of organ physiology), and obviously for necropsy and pathology.

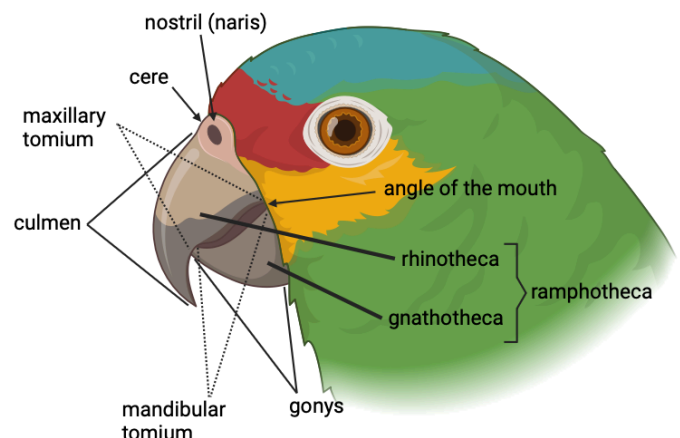
In this lecture, there will be a stronger emphasis on

integument and feathers, the respiratory system, and osmoregulation as they are the most clinically relevant, but also show the most differences from mammals.

Even though birds are not related to mammals [the clade sauropsid is actually a sister clade to the therapsids (group including the ancient therapsids and the current mammals)], it makes practical sense to focus and know the major differences from mammals as you are more familiar with mammalian anatomy and it may be easier for you to remember the anatomy and physiology this way.

TOPOGRAPHICAL ANATOMY

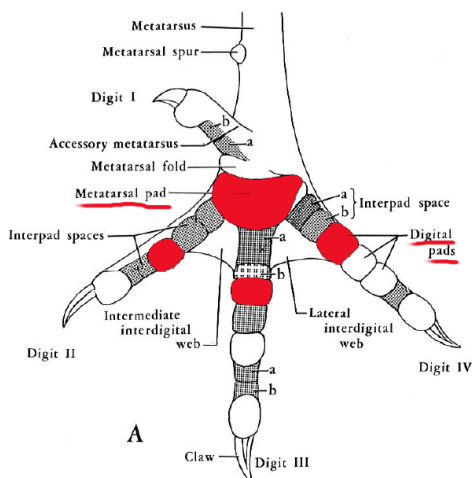
Topographical anatomy is important mainly to localize lesions and write in medical records. Most of the topographical terms are fairly intuitive and not bird-specific. The terms that you may not know are probably those of the head and face, especially regarding the beak. The top of the upper beak is called the **culmen** and the underside of the lower beak the **gonys**. The margins of the beak (cutting ends) are called the **tomium** (divided into the maxillary and mandibular tomium). These terms are important when describing beak fractures and beak overgrowth.



The beak is covered by a thick layer of keratin, called the **ramphotheca**. We divide the ramphotheca into the **rhinotheca (on the maxilla)** and the **gnathotheca (on the mandible)**. The beak is highly variable in birds and its shape reflects species adaptations to diets and ecological niches. The beak is especially strong in parrots. While the beak may have a general form in most birds, in some species, it is highly specialized such as the filtering beak of flamingos and the long beaks of water birds or some weirdos like crossbills and spoonbills. In some birds such as hummingbirds, the shape of the beak is directly related to the flowers they pollinize giving a striking example of co-evolution. Interestingly, toucans have a large beak mainly to help them thermoregulate (vascularization varies inside depending on the need for heat exchange).

On the head of some birds, there are ornaments or protuberances. For instance, hornbills have a casque (connected to the sinus in some species), mute swans have a knob at the base (more prominent in males), chickens have a comb and wattles, turkeys have a snood, and some birds also have feather ornaments such as the long erecting crest of cockatoos.

Another feature that tends to vary with specific bird adaptations is the avian foot. Some species have a purely cursorial foot such as ostriches (with only 2 toes), other ratites, and bustards while other species have a more perching foot such as raptors and most passerines. Raptors have long and sharp claws (talons) used for hunting and fishing. Water birds have webbed feet. The webbing may be continuous such as in Anseriformes or discontinuous around toes or even phalanges like in grebes and loons. On the plantar surface of the foot, there are some pads that cushion the foot and made of fat and connective tissue. The main one is the **central metatarsal pad** in the middle, which is frequently affected by bumblefoot in raptors.



Lucas & Stettenheim, Avian Anatomy: integument, 1972

The toe arrangement on the avian foot varies and is important to know.

Toe arrangement	Picture	Species
Anisodactyl		Most birds Hawks, eagles, falcons Canaries, most passerines
Zygodactyl		Parrots, cockatoos Woodpeckers Cuckoos Owls and ospreys are facultative zygodactyl
Heterodactyl		Trogon
Syndactyl		Most coraciiformes such as hornbills and kingfishers
Pamprodactyl		Most swifts

The wing shape also varies and depends on the flying strategies and biotope (complex such as wooded areas or lots of winds such as open space areas). The wing is characterized by its **aspect ratio** (wingspan divided by width) and its **wing loading** (weight divided by area). Soaring birds at slow speed have long and rounded wingtips (low aspect ratio) with emarginated feathers and low wing loading. Gliding birds in windy areas and fast birds have long wingspan with high aspect ratio and pointy wings (falcons, sea birds). Small birds with mainly a flapping type of flight have short and rounded wings.

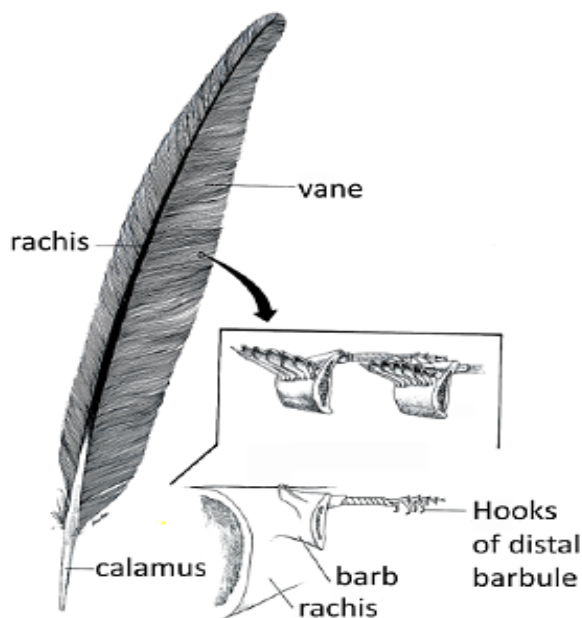
FEATHERS

Feathers are unique structures produced by feather follicles on the skin. They are undoubtedly the most complex structures produced by the skin of animals. Feathers have many roles, namely for **flight**, insulation (their original function), communication, and sensitive functions (for instance, birds sense air flows on their wings with specialized receptors at the base of filoplumes, and

may also have sensitive bristles on their face in a manner similar to cats).

There are different types of feathers you may find on a bird's body. **Contour feathers** are the most common and the archetypical feather. They are mainly used for flight and to cover the body (covert feathers). **Down feathers** are mainly used for insulation and are found under contour feathers. They include **powder down** feathers, typically found on the flanks, which produce a thin keratin dust used for plumage maintenance (for instance, it is very prominent in grey parrots). Bristle feathers are thick feathers mainly found on the face and used to protect facial features (such as eyes, mouth, and ears) from trauma. They are especially prominent in insectivorous birds and nocturnal birds. Filoplumes are tiny and simplified feathers essentially associated with other feathers and which primary role is sensitive.

A feather is made of a central shaft divided into the hollow and short **calamus** proximally and the long and full **rachis**. The rachis bears the vanes composed of a multitude of barbs. Each barb has proximal and distal **barbules**. The distal barbules have hooks that hook onto the grooves of proximal barbules of the adjacent barbs. This system allows the feathers to be light, absorb trauma easily, but also to provide extreme flexibility as the barbules and barbs slides within each other when the feather is bent.



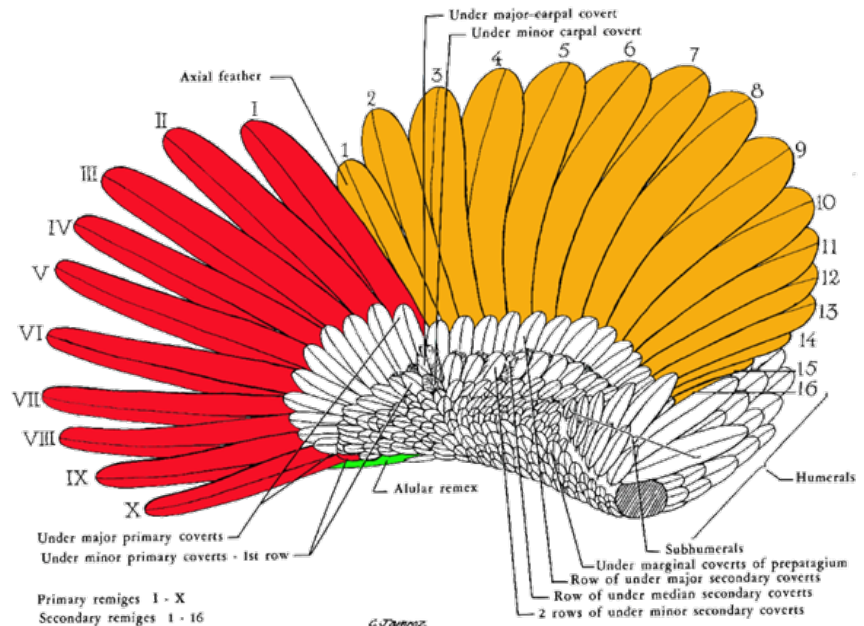
Lucas & Stettenheim, Avian Anatomy: integument, 1972

The hollow calamus is used clinically to do **feather**

imping [feather repair of a broken feather: a donor's feather (molted feather or from a dead bird) is cut at the level of the calamus, a piece of wood or fiberglass is placed inside the calamus/rachis, and then the distal end is introduced in and glued to the calamus of the broken feather of the recipient].

Tail feathers are called **rectrices** and are symmetrical.

Lucas & Stettenheim, Avian Anatomy: integument, 1972



Flight feathers are called **remiges** and are asymmetrical with the leading edge narrower than the trailing edge. Other feathers are simplified and have a short rachis with reduced and downy vanes.

On the wing, flight feathers are composed of **10 primary feathers** (primary remiges) and a variable number of **secondary feathers** (typically 20 or more, also called secondary remiges). **These feathers are attached to the periosteum of the bones** (carpometacarpus for the primaries and ulna for the secondaries) and are covered dorsally and ventrally by smaller covert feathers arranged in successive rows. They are numbered in a specific order (from base to wingtip). The **alula** (the bird's thumb) bears 4 alular feathers (or alular remiges). Most birds also tend to have **12 rectrices**, the 2 middle ones protect the other ones when the tail is folded and are called the deck feathers.

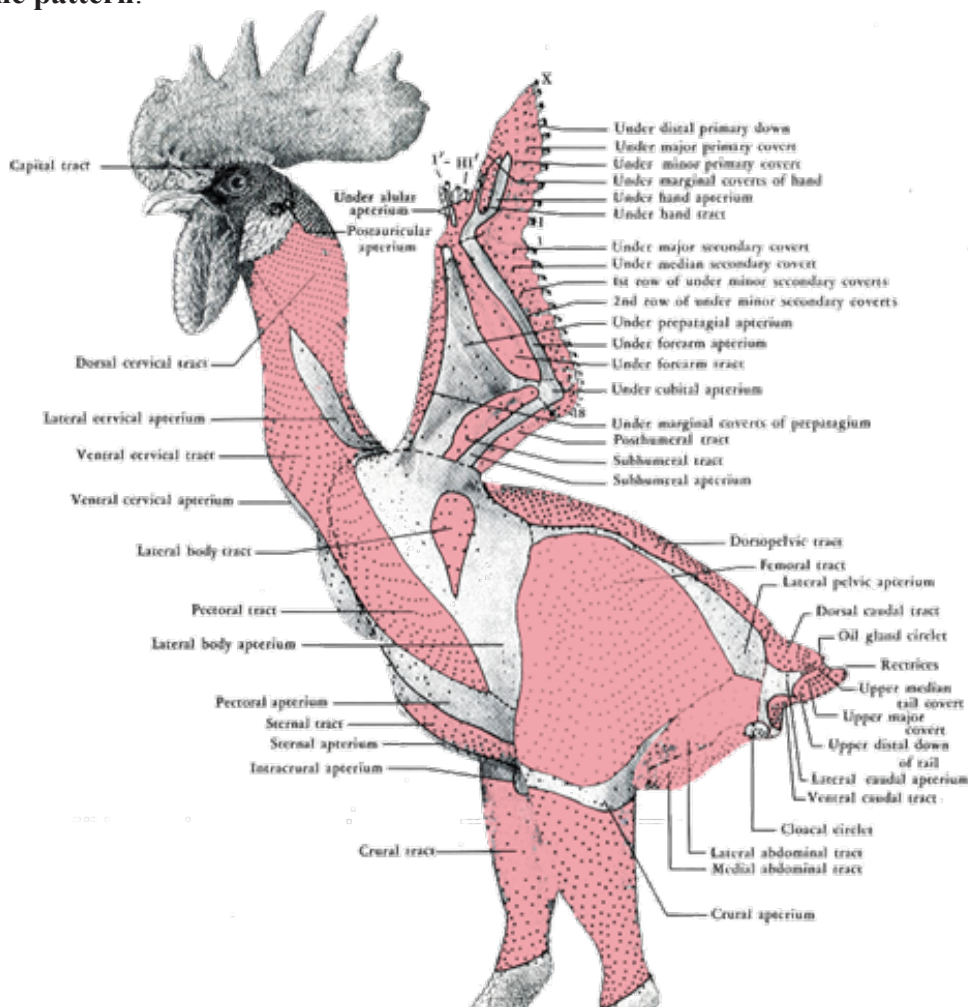
For the feathers to be evenly spaced and connected with each other when the wing is open or closed, they have to be linked by anatomical structures. The interremigial ligament connects all the wing feathers together and, if damaged, results in uneven spread. The ulnocarporemigial aponeurosis connects the proximal primaries together. In the body, most feathers are connected

by a complex network of muscles. Thus, birds can erect their feathers to increase the air layer over their skin for increased insulation (ptilorection, fluffed up bird).

The avian body is actually not entirely covered in feathers, with a few exceptions such as in penguins and some other sea and arctic birds. They are distributed in specific areas of the body called **pterylae** or **feather tracts**. The corresponding featherless areas are called **apteriae** (or **apterium**). The apterium can be used to our advantage for venipuncture for instance (lateral cervical apterium) or doing certain surgeries without the need for plucking.

Birds maintain their plumage in various ways as it is essential for flight, insulation, and interaction with conspecifics. **Most birds molt their feathers annually in a species-specific pattern.**

al to the tail feathers and called the **uropygial gland**. It is the only skin gland in birds and is bilobed. Birds use the tip of their beak and spread the oil over their plumage during the process of **preening**. Not all birds have a uropygial gland. The secretion may also be important for plumage waterproofing, but is not the only waterproofing mechanism of the feathers (most of the waterproofing is actually achieved by the specialized anatomy of the barbules of water birds). During preening, they also reorder their feathers and reconnect the barbs. Some birds also use water or dust bath for cleaning. Keratin dust produced by the down feathers may be important in some species.



Some birds molt several times a year for intraspecies communication reasons (mainly ducks with spring and eclipse plumages). Large birds only do partial molt and renew their entire plumage over 2 or more years. Ducks molt all their flight feathers at once and are essentially flightless during that time. Some birds may use the oil of a specific gland located on the rump, just crani-

Modified from: Lucas & Stettenheim, Avian Anatomy: integument, 1972

SKELETOMUSCULAR SYSTEM

The avian skeleton is characterized by two main features:

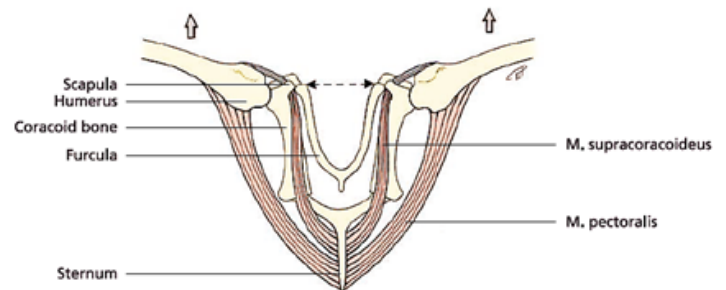
-Pneumatization of bones: most bones are hol-

low and infiltrated by diverticuli from the air sac system except for distal bones of the limbs. In some species, such as members of the orders peleciformes and cathartiformes (New-World vultures), the pneumatization is extreme. In other birds, such as diving birds and penguins, the pneumatization is reduced. Regarding the skull, the braincase and lower jaw bones are pneumatized by the tympanic cavity and not the air sac system while the upper jaw and soft tissues are pneumatized by the infraorbital sinus.

-Fusion of bones: in birds, bone fusion is common. The axial skeleton is mostly fused except for the neck vertebrae, one joint of the trunk, and some of the tail vertebrae. The thoracic vertebrae are all fused together in a structure called the **notarium** and the lumbar, sacral vertebrae, ischium, ilium, and pubic bones are fused into the **syndesmodium**. The caudal tail vertebrae are fused into the **pygostyle**, which supports the tail feathers. On the limbs, the tarsal and carpal bones are fused to adjacent bones. As a consequence, there are bone names such as the **carpometacarpus** and **tibiotarsus**.

also called the keel in birds.

The bones of the thoracic girdle are especially important to know. The thoracic girdle is composed of 3 bones, the **coracoid** (the pillar joining the wings to the thorax), the **clavicles** (both are fused into the furcula), and the **scapula**. Together, these bones form the **triosseal canal**, within which the **supracoracoideus muscle** tendon (major wing upstroke muscle) passes to insert on the dorsal humerus. As such, a fracture of the coracoid prevents the action of the supracoracoideus muscle and affected birds typically cannot lift their wings above their vertebral column.



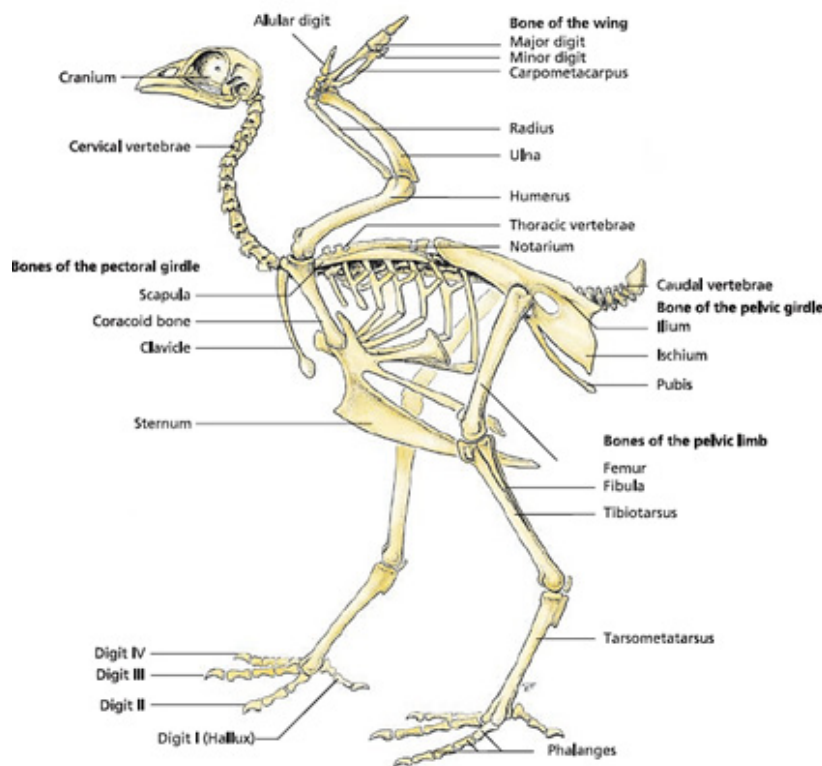
Koenig & Korb, Avian Anatomy, 2016

Birds only have 3 fingers on their hand, the major digit, the minor digit, and the alular digit.

A good rule of thumb for the number of phalanges of the toes is that it corresponds to the digit number +1 (so for the second digit, there are 3 phalanges). The toes are numbered from medial to lateral starting with the back toe (or the medial back toe in zygodactyl species).

The osteology of the skull is complex and is beyond the scope of this lecture. However, it must be known that birds have the ability to move their upper beak in a process called **cranial kinesis**. If cranial kinesis is achieved through elastic elements of the skull, it is called **rhynchokinesis** and if it is made possible by complete joints, it is called **prokinesis**. Parrots in particular have an **advanced prokinetic skull**. Upper beak opening is allowed by rotation of the quadrate bone at the base of the skull that pushes forward the palatine-ptyergoid complex forcing the maxilla upward. Beak injuries and fractures are common in birds and may occur at different level of their skull skeletal anatomy. Only CT scans can allow you to diagnose some of these fractures.

Knowing the muscular anatomy is not fundamental to the medicine of birds, but may be handy when performing surgical approaches to avian long bone fractures.



Koenig & Korb, Avian Anatomy, 2016

The sternum is very wide and plate-like and protects thoracic organs, especially the heart, which is completely enclosed within its internal surface (which limits diagnostics such as echocardiography for instance, but also limits cardiac compression during CPR). It is

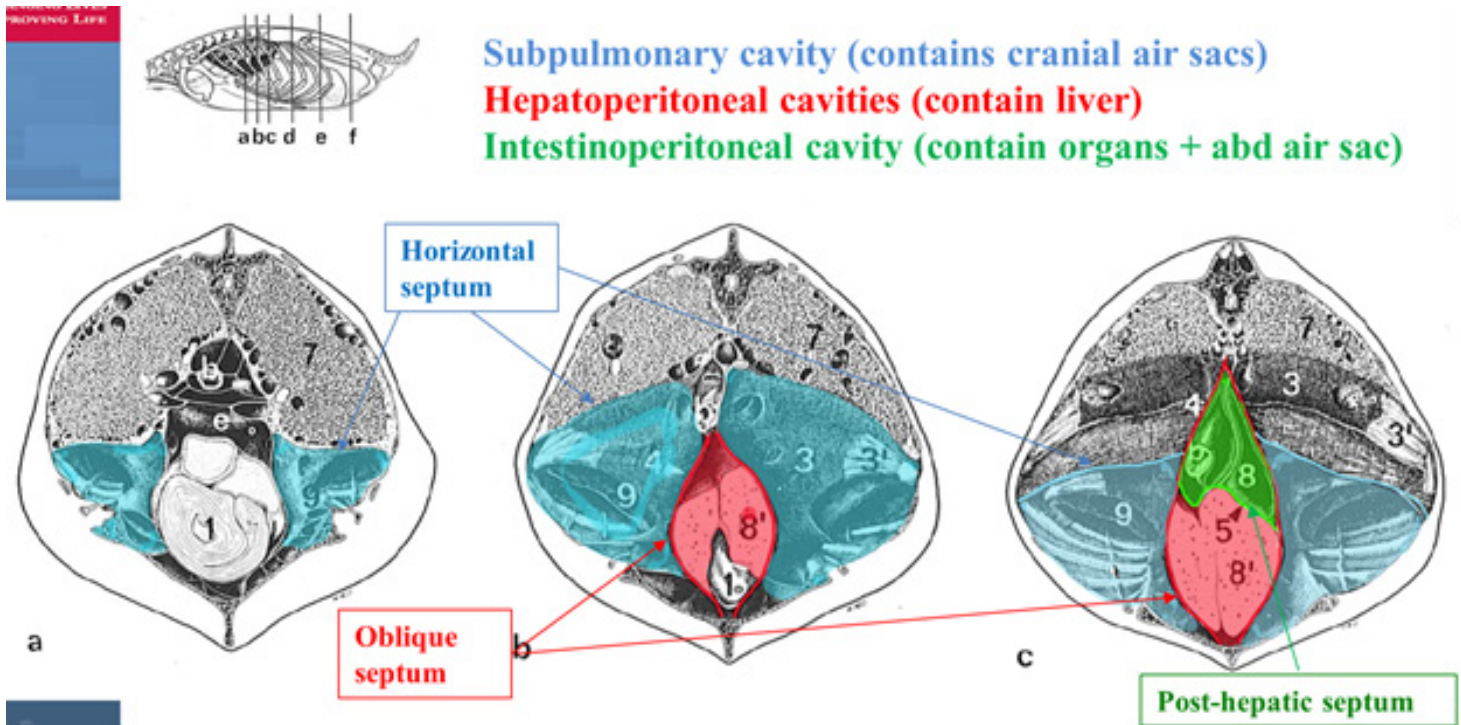
Intramuscular injections are typically performed in the large ventral pectoral muscles. There is also a small muscle and tendon at the leading edge of the wing called **tensor propatagialis**. In case of laceration of the patagium (the skin flap between the humerus and the forearm that makes most of the wing leading edge), that tendon needs to be sutured separately from the skin for functional repair.

COELOMIC CAVITIES

The coelomic cavities of birds are extremely complex and have evolved from structures found in reptiles. In particular, crocodilians have very similar body cavities than birds, being within the archosaur clade. The lungs are within the pleural cavities (not equivalent to mammalian pleural cavities, which are virtual cavities between pleurae). Below the lungs are two large cavities that have been created by the extending cranial group of air sacs; they are called the **subpulmonary cavities** and are not peritoneal cavities. They are separated from the lungs dorsally by the **horizontal septum**. These cavities surround cranially the pericardial cavity containing the heart and caudally the 4 **hepato-peritoneal cavities** (2 ventral and 2 dorsal, separated in the midline by the

ventral and dorsal mesentery and dorso-ventrally by the hepatic ligaments). The hepatic-peritoneal cavities are separated from the subpulmonary cavity by the **oblique septum** laterally. As such, birds do not have a mediastinal space. In the caudal abdomen, another large cavity is present and is called the **intestino-peritoneal cavity**, which contains most of the coelomic organs including the abdominal air sac (the only coelomic air sac) and with the exception of the liver (in the hepato-peritoneal cavities) and the kidneys (which are retroperitoneal). This large cavity is separated from the 4 hepato-peritoneal cavities by the **post-hepatic septum** (which, interestingly, is a functional diaphragm in crocodilians) and from the subpulmonary cavities by the oblique septum.

While it is complicated, it is still essential to know for surgery and diagnostic imaging (as peritoneal effusion may only be located in one of these cavities). If you do a skin incision in midline over the abdomen for instance, you end up in the intestino-peritoneal cavity. But you won't find the liver there, you will have to locate the post-hepatic septum cranially, incise it and hence find your way to the liver. If you make a mistake caudo-laterally, you may end up in the thoracic air sac system, which means you would have punctured the oblique septum and the thoracic air sac wall.



RESPIRATORY SYSTEM

The respiratory system of birds is truly unique and fascinating. It is the most efficient gas exchange system of any animal group. The system has many differences from mammals, which have direct consequences and implications for you in clinics (see below).

Major Anatomical and Physiological Differences in the Respiratory System of Birds Compared to Mammals

Fixed external nares
Larynx: lack of epiglottis and thyroid cartilages, no vocal cords
Complete tracheal rings
Elongated trachea, wider tracheal lumen (increased tracheal dead space)
Sound produced by the syrinx
Single paranasal sinus with many diverticulae
Lungs parenchyma made of parabronchi and air capillaries
Presence of large air sacs responsible for ventilation
Larger tidal volumes
No functional diaphragm
Fusion of parietal and visceral pleura
Active exhalation and inspiration
Respiratory system pneumatizes most bones and cavities
Thinner blood gas barrier
Cross-current gas exchange

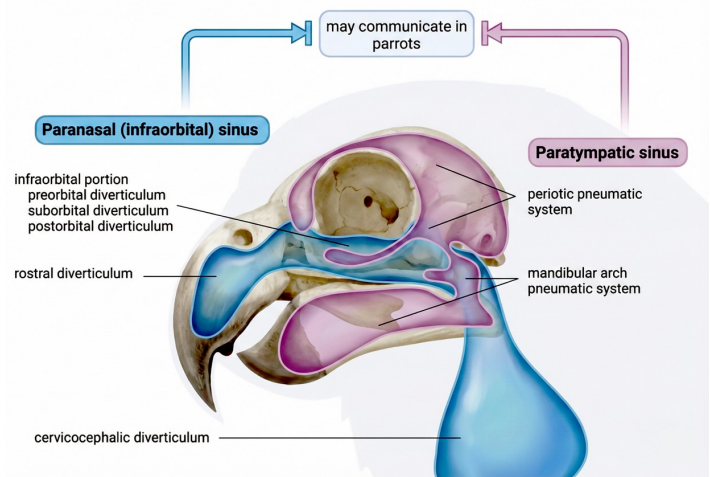
Major Clinical Implications of Some Anatomical Peculiarities of the Avian Respiratory System

Medical area	Implications
Anesthesia	-Use of uncuffed endotracheal tubes -Air sac perfusion anesthesia is possible -Increased respiratory depression of inhalants/inhibition of intrapulmonary chemoreceptors -Positive effect of IPPV on cardiac output, does not decrease blood pressure (unlike in mammals) -Fast induction and recovery -Increased tidal volume and tracheal dead space -Capnography not accurate with breach into the air sac system
Surgery	-Some surgeries require penetration into the air sac system -Wound irrigation may cause fluid aspiration (e.g. pneumatized bone fracture, coelomic surgery, sinusotomy)
Disease pathogenesis	-High susceptibility to airborne toxins -Poor drainage of sinusual exudate (dorsal openings into nasal cavities) -Respiratory diseases may spread to organs surrounded by air sac diverticula -Fluid, organomegaly, masses may decrease air sac volumes and thus ventilation
Diagnostics	-Air sac system allows coelioscopy without insufflation -Sinus exudate can be collected via transcutaneous punctures -Decreased ultrasound windows through the body, thus coelomic ultrasound is less useful in birds -Better delineation of coelomic organs on radiographs than in mammals

Major Clinical Implications of Some Anatomical Peculiarities of the Avian Respiratory System

Medical area	Implications
Therapeutics	-Nebulized particles should be below 1-2 um to reach air capillaries -Nebulized particles are less likely to be deposited into cranial air sacs -Low drug distribution into air sac membranes

The upper respiratory system is essentially composed of the nasal cavities and a single paired paranasal sinus, the infraorbital sinus. The infraorbital sinus is very large, pneumatizes most of the soft tissue and the maxilla and has many diverticuli. The largest diverticulum in parrots is the **cervicocephalic diverticulum** (not to be confused with the small cervical air sacs) that extends down the neck. Both sinuses communicate in parrots whereas they don't in passerines (so sinusitis may extend contralaterally or not depending on the species).



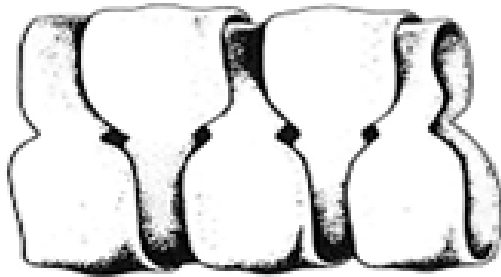
The lateral wall of the infraorbital sinus is made of soft tissue so it is easy to access for surgery or sinusual lavage for diagnostic purpose. However, the infraorbital sinus only has a dorsal opening to the nasal cavity, making it difficult for sinusual exudates to drain into the nasal cavity.

The choanae are located on the roof of the mouth and have **papillae** on their lateral margins. These papillae tend to become blunted with hypovitaminosis A, so it should always be examined during the physical examination.

The skull in birds is also pneumatized by the **paratympatic sinus** arising from the tympanic cavity. It extends into many diverticuli throughout the braincase and the mandible. In parrots, the two head pneumatizing system (paranasal and paratympatic) may communicate, but they typically don't in most birds.

The lower respiratory system starts with the glottis and trachea. The glottis of birds is simplified with a lack of epiglottis and vocal cords, making it relatively

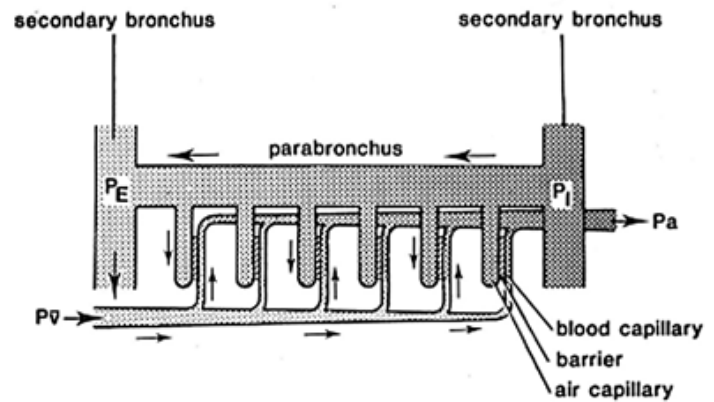
easy to intubate birds. In some birds, however, there is a central cartilage midline within the glottis called the *crista ventralis* (found in penguins, some sea birds, hornbills, pelicans and so on). The trachea has **complete and overlapping tracheal rings**. Consequently, birds should **never be intubated with cuffed tracheal tubes** as the trachea can be damaged since it cannot expand in diameter. Lots of birds have tracheal peculiarities such as extreme elongation and coiling (the prime examples of that are the trumpeter swan, the whooping crane, and the trumpeters), slits like the emu, or other.



King & McLelland, birds: their structure and function, Bailliere Tindall, 1984

The **syrix** is the avian second larynx and is located at the bronchial bifurcation. It is typically composed of specialized cartilages making kind of a sound box with vibrating membranes, the tympaniform membranes. In birds with extremely developed vocalization such as passerines and parrots, intrinsic syringeal muscles are present to modulate the voice. Male ducks typically have a large **syringeal bulla**, visible on radiographs (so you can sex some ducks on xrays).

From the syrix emerge two primary bronchi leading to the lungs. Birds have two non-expandable lungs that are only used for oxygenation. Ventilation is performed by the **air sac system**, which acts as bellows. Within the lungs, the airway becomes progressively smaller through successively the secondary bronchi (also called the mesobronchi), the **parabronchi**, and the air capillaries, the latter being the site of oxygenation. Within the very tiny air capillaries (an air capillary is about 100 to 200 times smaller than a mammalian alveola, making for greater respiratory surface), the air comes into contact with adjacent vessels in such a way that there is a **counter-current gas exchange** (much like in fish gills, but unlike in mammalian alveoli where air just equilibrates with the blood). This, along with thinner air-blood barrier and larger exchange surface, considerably increases air-blood gas exchange.



King & McLelland, birds: their structure and function, Bailliere Tindall, 1984

Thus, birds can uptake more oxygen and release more CO₂ into the air than mammals. That's why inhalant gas anesthesia is so fast in them as they uptake the anesthetic gas much faster than mammals (birds induce in seconds compared to minutes in mammals). Lungs do not participate in ventilation in birds. This function is performed by the air sac system. This system is composed of 2 functional units (with a total of 9 air sacs):

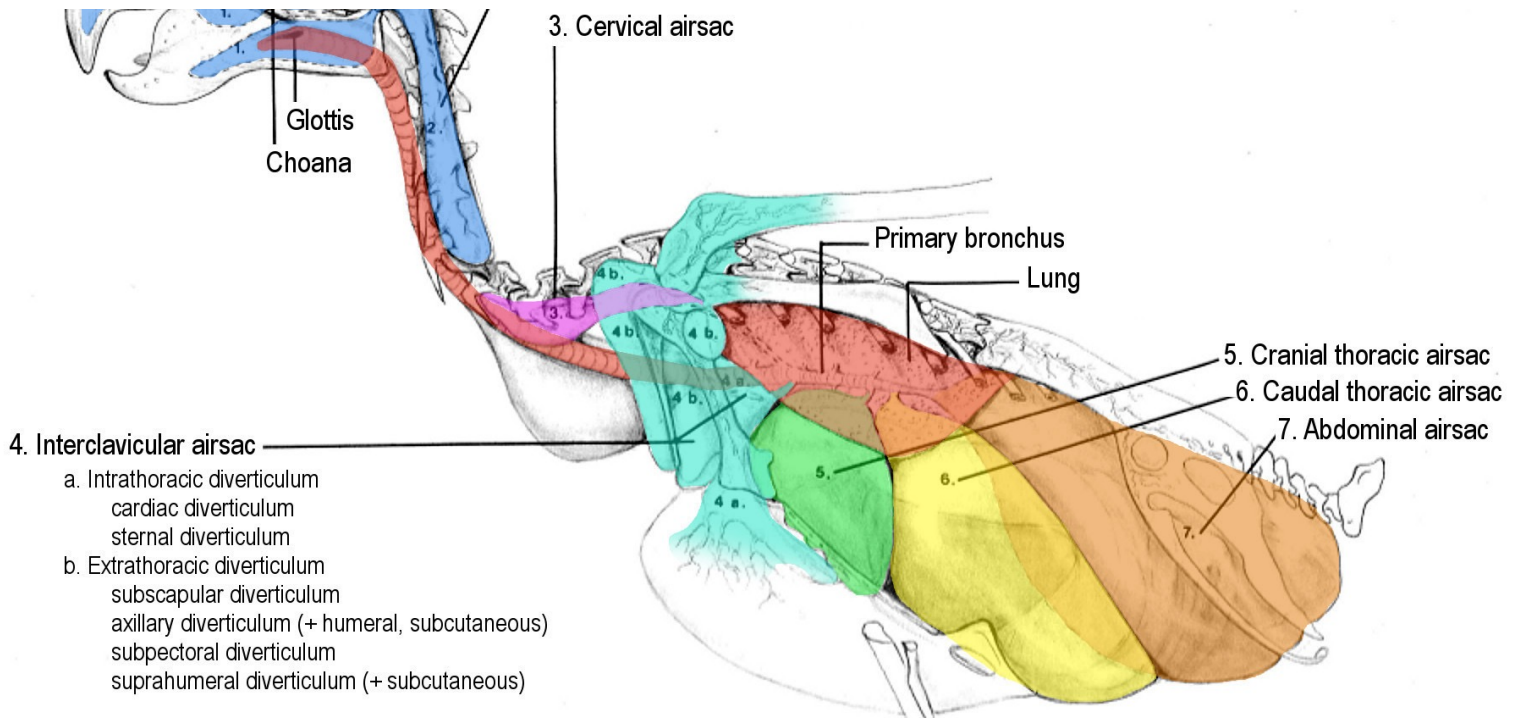
- The caudal group of air sacs composed of the paired **abdominal air sacs** and the paired **caudal thoracic air sacs**, which get **inspiratory oxygenated air**.

- The cranial group of air sacs composed of the paired **cranial thoracic air sacs**, the paired **cervical air sacs**, and the unpaired **interclavicular air sac**, which receive **expiratory deoxygenated air**.

Air moves from the lungs to the caudal group of air sac, then air sac compresses upon the action of respiratory body wall and intercostal muscles, which pushes the air in a unidirectional way through the parabronchial lungs. After gas exchange, the air ends up in the cranial group of air sacs, and is then expelled through the trachea. Thus, the **respiratory cycle of birds required 2 inspiration-expirations** for a complete air renewal in the system.

Air sacs are connected to the lungs by ostia. Air sacs have **many diverticuli and pneumatize most of the bird's body**. The interclavicular air sac, in particular, has diverticuli throughout the thorax, wings, and cranial subcutaneous tissues. The air sac system is not connected to the nasal cavity – sinus system.

In birds, both inspiration and expiration are active. The main ventilatory muscles are the intercostal muscles. When birds require a higher respiratory amplitude or with increased respiratory efforts brought about by respiratory diseases, they can recruit additional ventilatory muscles. Some of these muscles are tail muscles. Recruitment of these muscles results in tail-bobbing.



CARDIOVASCULAR SYSTEM

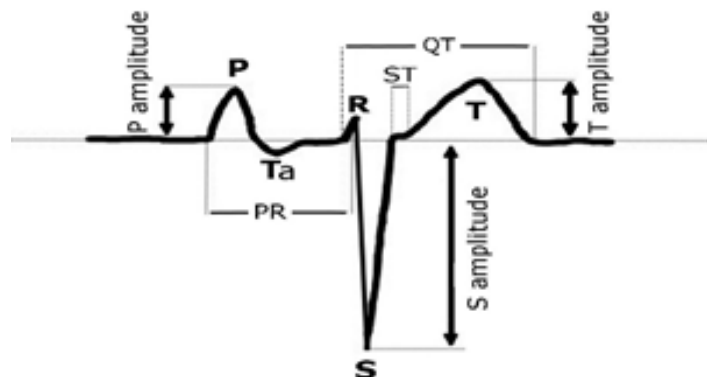
To keep up with a higher metabolism and a very efficient respiratory system, birds also have a high-performance cardiovascular system.



The avian heart is bigger, especially in falcons, and, as a consequence, has a higher stroke volume, heart rate,

and generates higher blood pressures in arteries. The heart of birds, as can be expected, is a four-chambered heart. While the left atrio-ventricular heart is similar to mammals, the **right atrio-ventricular valve** is quite unique. It is a muscular valve with its own innervation and comes from the lateral wall of the right ventricle. We don't talk about mitral or tricuspid valves in birds because of their different structures.

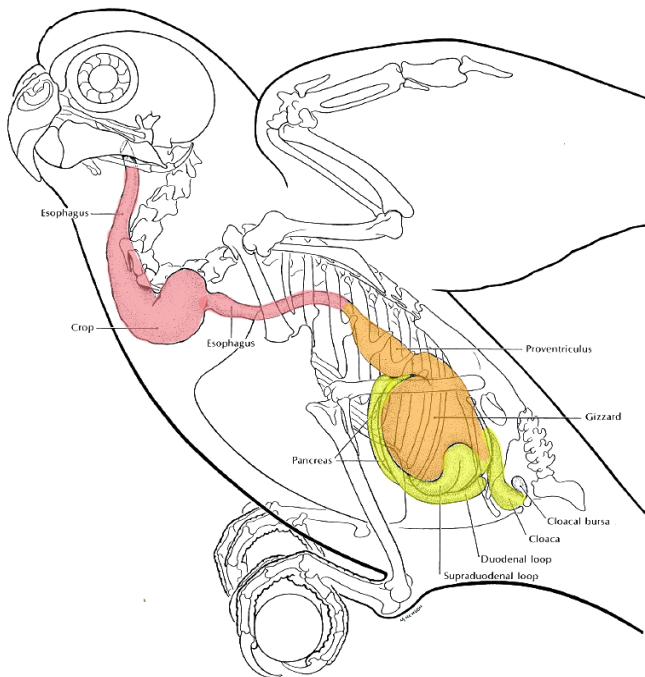
The ECG of birds is different than in mammals as they have an **"inversed" ECG**. It is caused by a different ventricular depolarization pattern resulting in a negative mean electrical axis (the MEA is mainly positive in mammals).



GASTROINTESTINAL SYSTEM

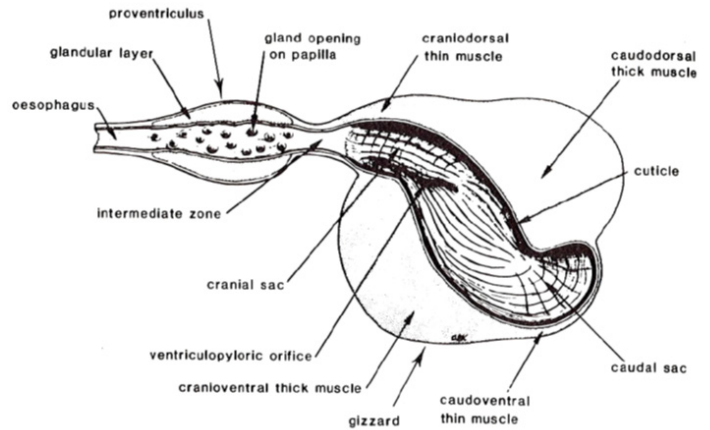
The esophagus of birds frequently has a blind pouch ventrally in the lower neck called the **crop**. It is essentially a storage structure and is present in most birds. Some of the birds that do not have a crop include owls and ducks. When force-feeding birds, the food is deposited into the crop or directly in the stomach for crop-less birds. With GI disease, it is common to have crop stasis.

The esophagus then opens into the **proventriculus**, the avian first stomach, which is mainly a secretory stomach. The second stomach of birds is called the **ventriculus** and is a grinding stomach. It has two pairs of muscles. In carnivorous birds, the ventriculus is similar to the proventriculus. The ventriculus is lined up by a hard layer of carbohydrate-protein complex, called the **koilin layer**. The junction between the proventriculus and the ventriculus is called the **isthmus** and is the preferential site of specific diseases (macrorhabdosis, adenocarcinoma).



Roskopf & Woerpel, Diseases of cage and aviary birds, Williams & Wilkins, 1996

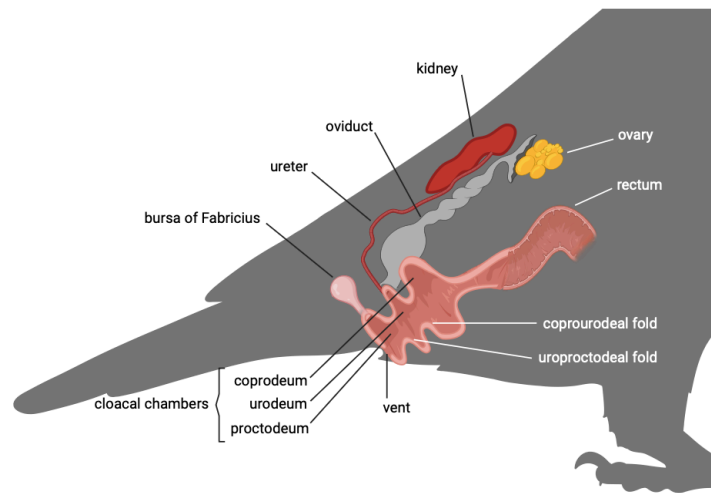
In granivorous birds, such as parrots and pigeons, there is a complex proventricular-ventricular cycle of contractions (see [video](#)). It is normal for the intestines to have retroperistalsis in the ventriculus. That cycle of contraction is disrupted with GI diseases and neurological diseases such as avian bornavirus infection.



King & McLelland, birds: their structure and function, Bailliere Tindall, 1984

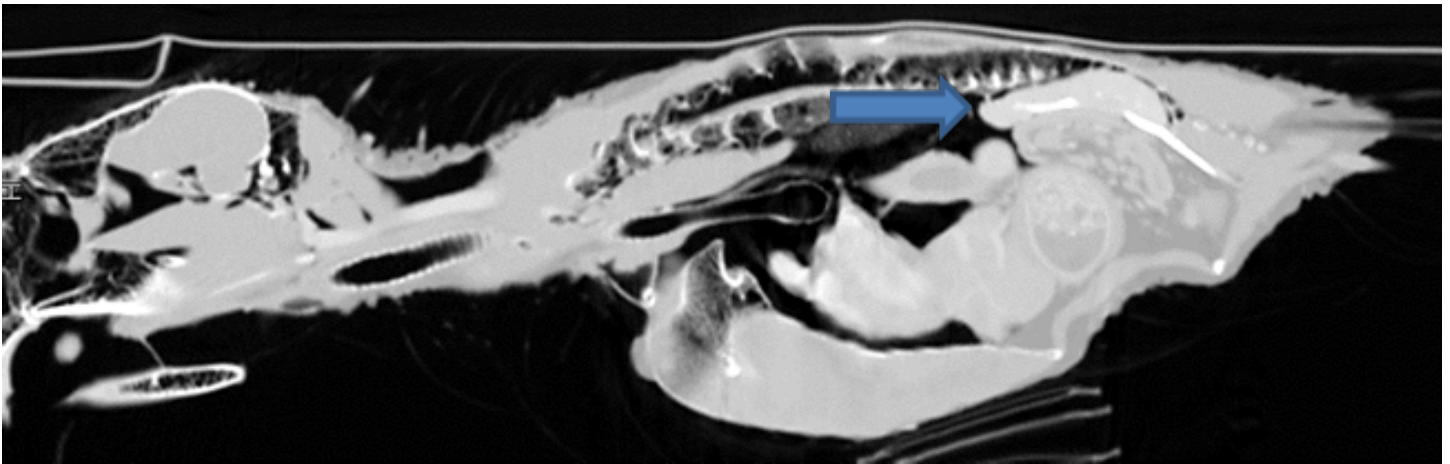
The ventriculus then opens into the duodenum. The pancreas is contained within the duodenal loop. Birds have both small and large intestines, and the demarcation between the two is marked by the **Meckel's diverticulum**, the remnant of the yolk sac. In the large intestines, birds have 2 **ceca** which can be large (e.g. ostriches, owls, chickens, turkeys) or vestigial (e.g. parrots, passerines).

The cloaca is the common end structure to the digestive, reproductive, and urinary system. It is made of 3 chambers being, from cranial to caudal, the **coprodeum** (where the rectum empties), the **urodeum** (where the reproductive tract and ureters empty), and the **proctodeum** (where the bursa of Fabricius is). The phallus of birds, for species that have one (ducks and ratites), is also in the proctodeum.



URINARY SYSTEM

The end-product of protein catabolism in birds is **uric acid and not urea**. It is produced in the liver. Uric

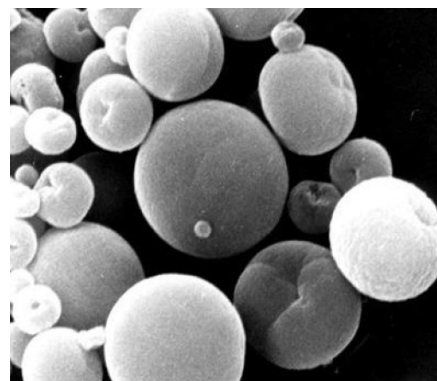


The avian kidneys are composed of several divisions (not lobes), generally 3, and are located within a ventral excavation of the synsacrum, dorsally. The kidneys are surrounded by abdominal air sac diverticuli, which make them usually inaccessible by ultrasound.

The renal vascularization of birds is unique in that a **renal portal system** is present that drains blood from the legs and the lower body. A valve is present, which controls the amount of shunting away from the system to the caudal vena cava. The role of this system is to increase tubular perfusion during dehydration to continuously excrete uric acid while decreasing glomerular filtration rate (to save water). Some important nerves from the lumbar and sacral nervous plexi run through or between renal lobes. That's why **renomegaly can lead to lameness** such as in budgerigar renal tumours. The kidneys excrete urine and uric acid into the ureters, which then empty into the urodeum. There is no bladder in any bird species (no bladder in ostriches too).

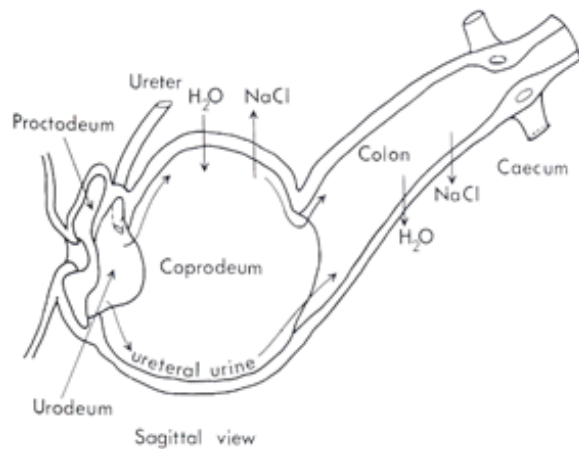
The avian kidney is not organized in cortex and medulla like in mammals, but into tiny renal lobules. Within lobules there are a myriad of nephrons. Birds have two types of nephrons. About 30% of the nephrons are **looped "mammalian" nephrons**, which have a loop of Henle and have great concentrating abilities, and the majority, about 70%, are **loopless "reptilian" nephrons**, which do not have a loop of Henle and have limited concentrating abilities. Consequently, the urine produced by the kidneys is less concentrated than in mammals. However, birds still have amazing water conservation mechanisms with being **uricotelic** and having **post-renal handling of urine**.

acid, while requiring more energy to produce, is also less toxic than urea, does not diffuse through membranes (which is handy in an egg), and has decreased solubility. Hence, birds require less water to eliminate nitrogenous waste than mammals. However, the drawback of uricotelism is that uric acid can precipitate at increased blood concentration, which will lead to irreversible **gout**. Despite that fact, birds are still able to excrete uric acid in massive quantity and in supersaturated concentration in urine. How do they do it? They actually package the uric acid within **muccopolysaccharide microspheres** also containing electrolytes such as sodium and potassium, which allows a higher concentration of uric acid in suspension.



Scanes, Sturkie's avian physiology, Academic Press, 2015

On top of that, urine, once excreted into the urodeum, flows back into the coprodeum, rectum, and ceca, where most water and electrolytes are reabsorbed (**post-renal handling of urine**) and where precipitation does not matter anymore. As such, uric acid, urine, and feces are excreted altogether when the bird is defecating.



Whittow, Sturkie's Avian Physiology, Academic Press, 2000

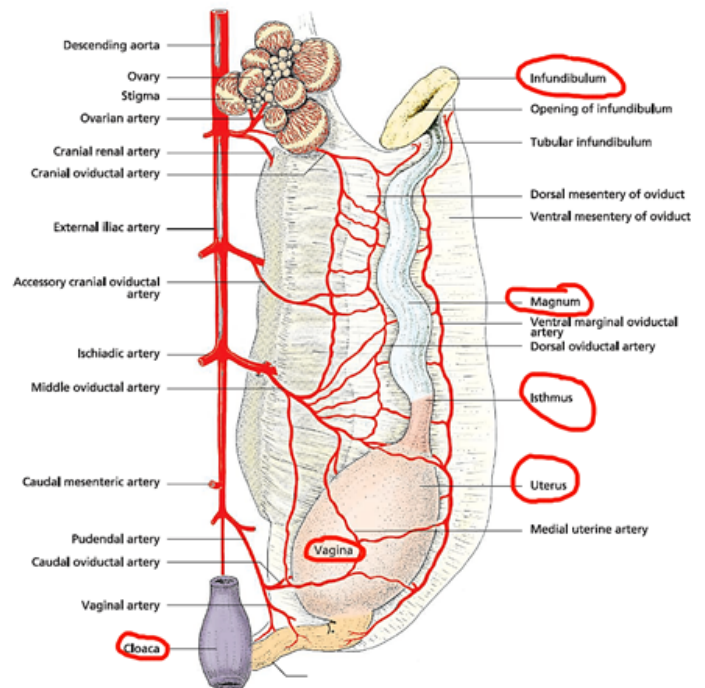


Birds also have the ability to regulate their GFR, while it is kept constant in mammals, and regulate their urine output depending on water availability. Some birds have salt glands, which can be the main osmoregulatory organs in them. Salt glands are supra-orbital glands and excrete salt mainly, allowing seabirds to drink sea water. The rate of electrolyte exchange in salt glands is astonishing (but still superseded by the rate of calcium movements in the bird uterus).

When taking all these mechanisms together, birds are great at conserving water.

REPRODUCTIVE SYSTEM

Female birds only have the left reproductive system with few exceptions (the most notable one being the kiwi). The oviduct is composed of 5 parts with specific function, the **infundibulum** (picking up oocyte, site of fertilization, deposition of chalazae), the **magnum** (deposition of albumen), the **isthmus** (deposition of shell membranes), the **uterus** (or shell gland, where the shell is made), and the **vagina** (sperm storage, laying muscles, where oviposition occurs). An egg takes about 24h to be produced.



Koenig & Korbel, Avian Anatomy, 2016

Egg-laying is a metabolically demanding process and lots of diseases are related to the female reproductive physiology for this reason. Prior and during laying, birds store calcium into their bones so it can be mobilized quickly for the shell. It is visible radiographically and is typically a good indicator of the reproductive status of a female bird, it is called **polyostotic hyperostosis**.



Before being laid, the egg assumes a specific position called **oviposition** whereby it has its small end pointing caudally and is in the vagina. Oviposition and laying are under hormonal control, mainly by **arginine vasotocin** (and not oxytocin like in mammals) and some

prostaglandins (E2 and F2 α). It is important to know because some drugs act on these (like nsaids).

Once the egg is laid, it needs to be incubated and turned, unlike in reptiles. It can be performed naturally by the bird obviously, but is also frequently performed artificially by breeders to increase production (and also because incubation inhibits ovulation, under the control of prolactin). Incubation also inhibits molting. This is why when you treat a chronically broody or hormonally active bird with hormones (such as deslorelin), they can lose a lot of feathers at once. Avian **incubation** is technically challenging and lots of species-specific parameters (mainly the humidity, temperature, and amount and frequency of turning) need to be closely maintained and monitored or diseases will arise (whereas reptile eggs don't need to be turned). The incubation period of most birds is 24-32 days.

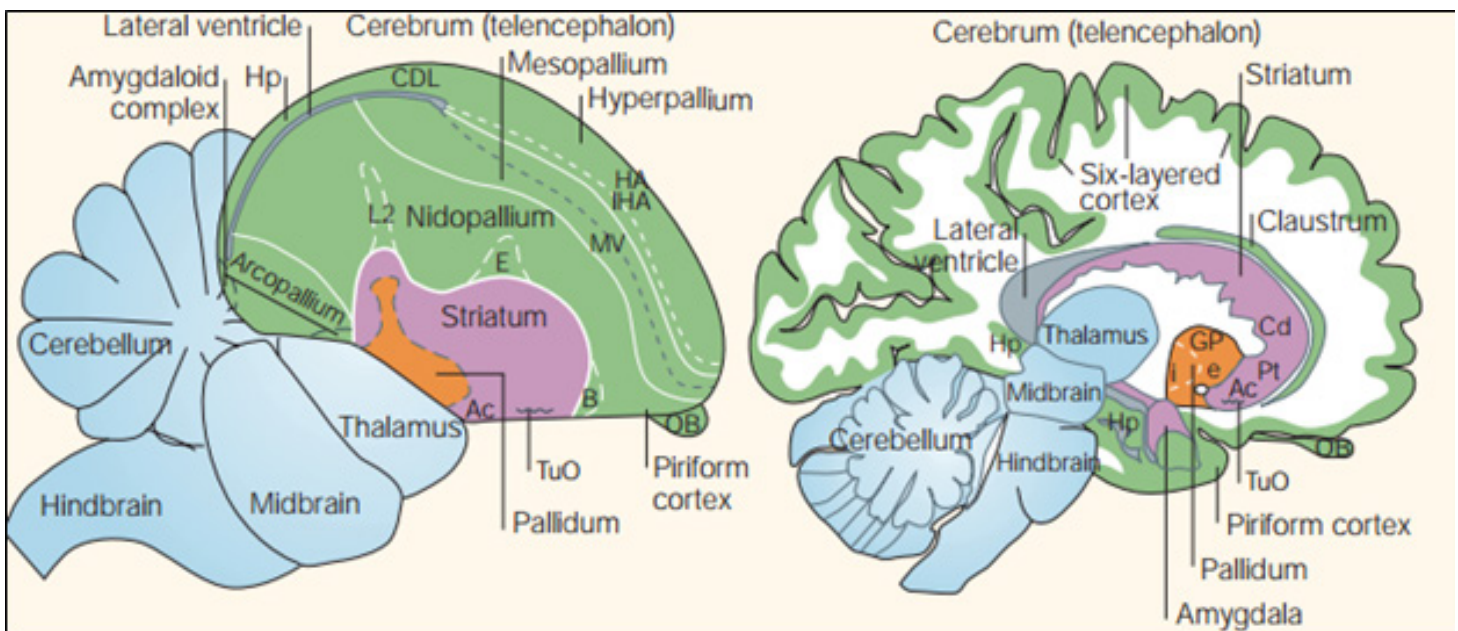
Males have right and left reproductive systems. Testicles are internal in birds, differently from mammals (except in marsupials) and can get huge in some species during seasonal reproductive activity (e.g. in pigeons and ducks, but usually not in parrots). The males of select species (ratites, ducks) have a phallus in the proctodeum that has lymphatic erectile mechanism.

Birds, as everybody knows, are among the smartest animals on the planet. Consequently, the brain is large comparatively to reptiles, amphibians, and fish and generally of similar size than in mammals.

Parrots and corvids, being the smartest, have the largest brains of all birds. The avian brain is also better organized than in mammals and birds have comparatively more neurons and a nuclear brain organization (instead of a layered organization within a cortex, as in mammals). Reptiles have a similar brain organization as birds. The equivalent structure to a mammalian cortex is called the **nidopallium** and is deep within the brain. Birds are very visual and consequently have very large optic lobes.

As in reptiles and mammals, birds have 12 cranial nerves. However, unlike mammals, the spinal cord extends to the limit of the vertebral column and birds do not have a cauda equina (so the neurological spinal localization is similar to the vertebral localization, unlike mammals, see your neuro lectures). Interestingly, birds have a “**glycogen body**” in the middle of the lumbar spinal cord, effectively splitting the spinal cord in two. It is readily visible on MRI and has confused more than one radiologists not knowing bird anatomy. Its function is debated, but it seems to be related to bipedal locomotion and shock absorption.

NEUROLOGIC SYSTEM



LYMPHOID ORGANS

I won't spend too much time on lymphoid organs. Basically, birds do not have lymph nodes and the liver performs this function instead. Other lymphoid organs include the spleen, the thymus (which regresses in adults, located along the neck), the **bursa of Fabricius** (where the name of B-cells come from, located on the roof of the proctodeum, involutes in adult birds), and diffuse organ-based lymphoid tissue (GALT, BALT...).

THE EYE

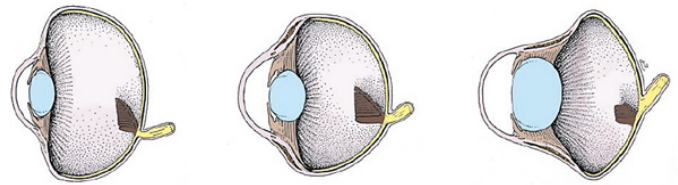
Within the special sense organs, I will only talk about the eye as birds are very visual animals. However, some birds have an exceptional sense of smell (e.g. Turkey vultures, kiwis) or an incredible hearing (e.g. owls). Some birds are almost blind (e.g. kiwis, oilbirds), but rely on other senses (smell in kiwis and echolocation in oilbirds).

Birds have an incredible vision with far better visual acuity than humans (and most mammals). Three globe shapes are commonly encountered in birds, the flat shape (chickens), the globose shape (most birds), and the tubular shape (owls).

The main differences from domestic mammals, which also make the avian optical system more efficient, are the following:

- Bigger eyes
- Striated iris muscles (so you can't dilate their pupils with atropine, but you can with curares)
- Presence of scleral ossicles (to protect the globe, which in most birds is extracranial)
- Avascular retina
- Presence of a pecten** (large structure in the fundus that flickers and provide nutrients to the retina and vitreous body)
- Presence of foveas (areas of maximum visual acuity in the retina)
- Minimal motions in eye sockets (birds compensate by exceptional head movements and balance)

Birds also have a pentachromatic vision and see in 5 colors including the UV (humans have a trichromatic vision).



Koenig & Korbel, Avian Anatomy, 2016

The Ten Most Important Facts

1. Birds have different toe arrangements on their feet (parrots are zygodactyl; passerines, hawks, and falcons are anisodactyl; owls are facultative zygodactyl).
2. There are different types of feathers on the avian body distributed in pterylae (feather tracts). Featherless areas are called apteriae. The jugular apterium is used for jugular venipuncture.
3. The avian skeleton is characterized by pneumatization and fusion of bones.
4. Birds have more complex coelomic cavities than mammals such as their 4 hepato-peritoneal cavities and 1 large intestino-peritoneal cavity.
5. The respiratory system is composed of non-inflatable lungs and 9 air sacs organized into 2 functional units (cranial group and caudal group of air sacs). The trachea has complete tracheal rings.
6. The heart is large and has a muscular right atrio-ventricular valve. The ECG has a negative mean electrical axis.
7. Some birds have a crop, all birds have 2 stomachs (proventriculus and ventriculus) and the cloaca has 3 chambers (coprodeum, urodeum, proctodeum).
8. Birds are uricotelic and have mechanisms for post-renal handling of urine.
9. Laying birds frequently have polyostotic hyperostosis. An egg about to be laid should be in proper oviposition.
10. Birds are smart with a brain differently organized than mammals (nidopallium vs cortex).